Calculus done right

first edition

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Introduction

Calculus is one of those subjects in mathematics where it is generally associated with immense difficulty and boredom to the average person. Many are afraid of taking on the challenge to learn calculus. However, it cannot be understated how important calculus is to the modern science. It is basically the *language* of the modern scientific world. Without calculus, we would not have the mathematical tools to describe continuum dynamics, quantum mechanics or even find the distance travelled by a object with non-constant acceleration. The goal of this book is to equip you with the proficiency to wield this important language which describes the world. However, while the book focuses on both the theory and application, it will not be the most rigorous text. Some parts may lack mathematical rigour in favour of more intuitive explanations. Hence, I will recommend you to read up other great resources on calculus by others. Lastly, it is important that you practice calculus, do the exercises and I assure you you will gain a deep understanding of the text. Happy reading.

PART

I

Precalculus

SECTION 1

Functions

SUBSECTION 1.1

Function basics

Before we start on the calculus topics, this part of the book will acquaint you with the mathematics you would need to understand in order to read this book. If you are familiar with the these topics, you may skip to the Part II on Calculus I. For the rest of you, the pace of this part will be very fast, it is recommended that you read up on concepts you do understand very well. We will start with the idea of a *function*. By now, I am sure that many of you may have seen equations like these:

$$x^2 - 5x + 6 = 0 \tag{1.1}$$

$$sin(x) = 0.5, 0 \leq x \leq \frac{\pi}{2} \tag{1.2}$$

$$y = mx + c \tag{1.3}$$

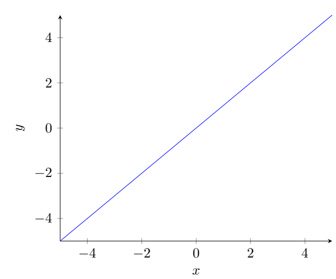


Figure 1. Graph of y=x.

Out of the three equations, notice that 1.3 is special. There is not some fixed x values that you must plug in. Instead, the y value varies with the x value. Given a x value, the equation "produces" a y value. This brings us to the idea of a function. A function is simply something that takes in a value, and "produces" another value based on the input. You may have came across functions before, such as the trigonometric functions like sin and cos. The value of $sin(x)$ varies with different values of x. Of course, it is time for me to give you the definition of a function.

Definition 1 A function f is a mathematical object that assigns its inputs, or *domain*, to its outputs, or *range*. The notation $f : A \to B$ means that the function has a domain A and range B. The notation $f(a)$ where $a \in A$ denotes the element of B to which a is assigned to.

Notice that A and B are sets.

Example Here are some examples of functions:

$$f(x) = x + 1 \tag{1.4}$$

$$f(x) = sin(x) \tag{1.5}$$

$$f(x) = x^2 - 5x + 6 \tag{1.6}$$

To evaluate the output of a function, you simply substitute the input to the expression that defines the function. Notice that for 1.4, it is simply the line $y = x+1$. Each value of $f(x)$ for some $x \in \mathbb{R}$ is just y. Take $x = 1$ for example, $f(1) = 1+1 = 2$ and for the line $y = x + 1$, notice the y value here is also 2. The function assigns each x value to $x + 1$. Hence, in this case, $f : \mathbb{R} \to \mathbb{R}$. For 1.5 , the function takes any real x value and maps it to $sin(x)$. As a result, its domain would be $\mathbb{R}$ and its range would be $[-1, 1]$. Again, it is just like the graph $y = sin(x)$. Lastly, the quadratic

1.6 is another real function(i.e. it has a real domain and range). Think about what happens when $f(x) = 0$, what value of x must you plug in to make $f(x) = 0$. Well, isn't it just the roots of the quadratic. Hence, the roots of a function are just the values x such that $f(x) = 0$. In this case, it would be 2 and 3. Thus, $f(2) = f(3) = 0$.

Actually, functions can be more than the usual $f(x) =$ *something*. Let's say you want to have a function that outputs "1" if the input is rational and "0" if it is not, you can define it as such:

This is actually the famous Dirichlet function that shatters certain intuitions on functions and will be discussed in later chapters.

$$f(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q} \\ 0, & \text{if } x \notin \mathbb{Q} \end{cases} \tag{1.7}$$

Here, $f(\frac{1}{2})$ would be 1 but $f(\sqrt{2})$ would be 0. This is an example of having conditionals in a function. Lastly, functions may take more than one input and produce more than one output, examples include:

$$f(x, y) = x^2 + y^2 \tag{1.8}$$

$$f(t) = (cos(t), sin(t)) \tag{1.9}$$

1.8 is an example of a *multivariate* function, in other words, it takes in multiple inputs. If $x = 2$ and $y = 2$, then $f(2, 2) = 8$. On the other hand, 1.9 does not have a single output. Instead, it has a tuple as its output. Evaluating it at $t = \pi$, notice that the output is $(-1, 0)$. What is interesting is that if you trace out its outputs on a plane for all $t \in \mathbb{R}$, you will get:

As you will see in Part IV of the book, this is the parametric equation of a circle.

A tuple is basically a set where order matters. You can treat it as if it is a coordinate.

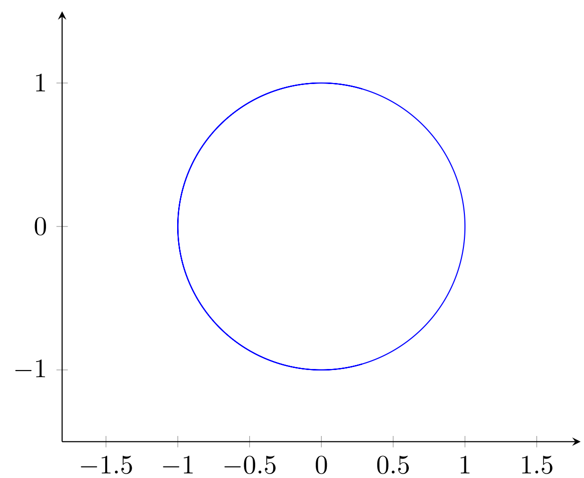


Figure 2. Plot of 1.9 for all $t \in \mathbb{R}$

Functions are the building blocks of mathematics, and this book will deal extensively with it. Therefore, it is paramount that you get the basics right. Try the following exercise:

Exercise

1. Given that $x = 2$ and $y = \pi$, evaluate $f(x)$ for:
   (a) $f(x) = x^3 + 1$
   (b) $f(x) = 2^x$
   (c) $f(x) = \begin{cases} x^4 + 1 & \textit{if } x < 2 \\ 3^x & \textit{otherwise} \end{cases}$

(d) $f(x,y) = \frac{sin(y)}{x}$

2. Given the following functions, write down $f : A \to B$ where A is the domain and B is the range(do not consider complex inputs and outputs):

   (a) $f(x) = x$

   This function has a special name, the ***identity*** *function. Notice how its input and output are the same.*

   (b) $f(x) = \frac{1}{x}$

   (c) $f(x) = \sqrt{x}$

3. Find the root(s) of the following functions:

   (a) $f(x) = x^2$

   (b) $f(x) = 2^x - 1$

   (c) $f(x) = sin(x)$

Solution For (1), simply plug in the x and y values. Hence, the answer for 1(a) is 9, 1(b) is 4, 1(c) is 9 and 1(d) is 0. On the other hand, for 2(a), notice that there are no restrictions on the domain and range, hence the answer for 2(a) is $f : \mathbb{R} \to \mathbb{R}$. However, 2(b) is not defined at $x = 0$, hence the solution is $f : \mathbb{R} \setminus \{0\} \to \mathbb{R} \setminus \{0\}$. Similarly, the square root is not defined for negative numbers(not considering complex numbers), therefore the answer for 2(c) is $f : \mathbb{R}_0^+ \to \mathbb{R}_0^+$. Lastly, for (3), simply find the x value such that $f(x) = 0$. The answer is as follows, 2(a) is 0, 2(b) is 1 and 2(c) is $n\pi$, $\forall n \in \mathbb{Z}$.

Subsection 1.2

Function composition and inverses

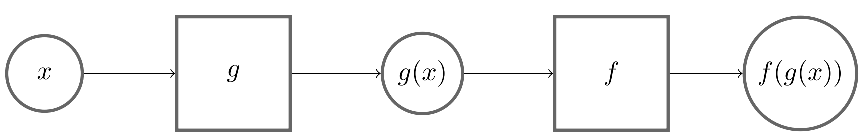
In this chapter, we will look at composition of functions and inverse functions. You can use the normal arithmetic operations on functions like how you would to numbers:

Example If $x = 2$ and $f(x) = x^2 + 4$:

$$f(x) + f(x) = 2^2 + 4 + 2^2 + 4 = 16$$

$$\frac{f(x)}{2} = \frac{2^2 + 4}{2} = 4$$

What is more interesting, perhaps, is taking the *output* of a function and use it as the *input* to another function. You may also think of it as "piping" one function into another. This is called *function composition*.



Note that the rectangles are functions and circles are values.

Figure 3. Visual representation of function composition.

Another notation for composing functions together(which we will not really use) is as such:

$$f(g(x)) = (f \circ g)(x)$$

When you compose the same function n-times, the notation for it is:

$$f^n(x) = \underbrace{f(f(\ldots f(x)))}_{n-times}$$

However, this does not apply to trigonometric functions. $sin^2(x)$ is $sin(x) \times sin(x)$ instead of $sin(sin(x))$ and $sin^{-1}(x)$ is $arcsin(x)$.

Example Here are some examples of composing functions together: If $x = 3$ and $f(x) = \pi x$, $g(x) = sin(x) + 1$:

$$g(f(x)) = sin(\pi \times 3) + 1 = 1 \tag{1.10}$$

$$f^3(x) = \pi^3 x \tag{1.11}$$

There is nothing much here, really, just evaluate the function from the innermost function to the outermost.

Theorem 1 Given functions $f : X \to Y$, $g : Y \to Z$ and $h : Z \to W$, prove that the composition of the three functions is *associative*, in other words:

$$h \circ (g \circ f) = (h \circ g) \circ f$$

PROOF Let $x \in X$, we have:

$$h \circ (g \circ f(x)) = h(g \circ f(x)) = h(g(f(x))) = h \circ g(f(x)) = (h \circ g)f(x)$$

This proves the associativity of function composition. QED

Remark This is just a fun little exercise for the reader and ultimately is not that important to the entire book, just a good identity to keep in the back of your mind.

Next, we will delve into inverse functions. Let us start with one I believe you may have seen before, look at the following figure:

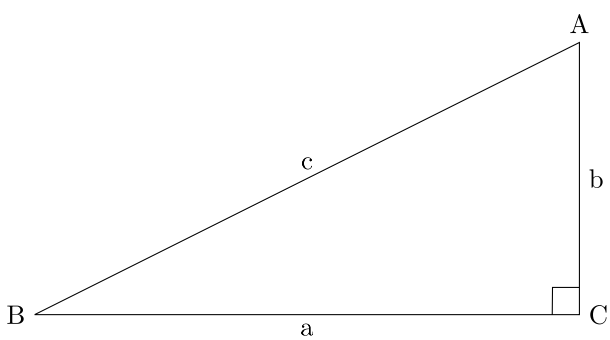


Figure 4. Figure of right triangle ABC.

If I ask you what is $arcsin(B)$, you may say it is $\frac{b}{c}$. But, what is $arcsin(sin(B))$, well, that is just B, isn't it. Hence, the $arcsin$ sort of "cancels" the sin. We call $arcsin$ the *inverse function* or just inverse of sin. Generally, given a function f, f^-1 denotes the inverse function such that $f^-1(f(x)) = x$. Notice that if we apply both of the functions together, we get the identity function, that is:

$$f \circ f^-1 = id_x \textit{ where } id_x \textit{ is the identity function} \tag{1.12}$$

The identity function is just $f(x) = x$

To find the inverse function of a function f, we have to find the function that maps the outputs of the f back to its inputs. To solve algebraically, we set $f(x)$ for some arbitrary x to y, and make x the subject of the equation.

Example Given that $f(x) = \frac{x+7}{3}$:

$$\begin{aligned} f(x) &= y \\ \frac{x+7}{3} &= y \\ x+7 &= 3y \\ x &= 3y-7 \end{aligned}$$

Hence, the inverse function $f^-1(y)$ is $f^-1(x) = 3y - 7$.